LEDGER-CORE · PART 2

# Architecture & Trade-offs

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The core has one primitive: `balance(account, value_date_upto, as_of_posting_day)`, a sum over an append-only log where every record carries the day it was posted and the day it is effective. There is no running balance. Hold state, the once-per-day fee check and the interest schedule are all queries over that log. That is why it is correct under backdated entries, and why it is slow.

## 1 Append-only at scale

### What breaks first: the end-of-day close

Every close re-checks every past day for every account, because a backdated entry can make an already-closed day negative. Each check scans the whole log. One close costs about *days × accounts × entries*, and entries grow with *days × accounts*: quadratic in both. At 100× accounts it does roughly 10,000× the work, and the close is the one job with a deadline. Nothing goes wrong; it just stops finishing on time.

### Where state grows without bound

The three logs (entries, authorizations, errors) grow by design: they are the audit record. The set of seen event ids grows by accident. Every lookup is a linear scan, and one global sequence number couples all accounts, making the engine a single writer.

> **Cheapest fix: close the period**
>
> Pick a horizon *H*, the last closed business day. Refuse any entry value-dated before *H* - one comparison in `append_entry`. At each close, write one snapshot row per account: balance, holds and fee flags as of *H*.
>
> Then every balance is snapshot plus a scan of entries after *H*; the close re-checks only the days since *H*; the id set only needs ids after *H*. Audit is unchanged because the snapshot is derived from the log. It bounds the problem instead of speeding up the scan.
>
> Left for later: the log stays unbounded (five-year retention), and parallel processing needs a sequence per account. An index comes after those.

## 2 Value-dated entries in production

A value-dated entry rewrites history. In Part 1 one backdated debit restated three closed days, produced three fees in one close and declined an authorization; its reversal restated the days again and left the fees standing. In a UAE-licensed bank each effect has an owner.

| | |
|---|---|
| **Customers and complaints** | Issued statements become wrong, and a fee appears for a day the customer saw as positive. CBUAE Consumer Protection Standards (Circular 8/2020, Art. 8): acknowledge a complaint within 2 business days, answer within 30, keep the file five years. Re-assessed fees will be the main source of those complaints. |
| **Fees** | Overdraft charges sit under the CBUAE retail fee caps (Regulations 29/2011, Annexure 2, amended 2018); a new fee or a change above 5% needs prior CBUAE approval. An engine that turns one backdated debit into three charges is |

| | |
|---|---|
| | a fee behaviour compliance must approve and the customer agreement must describe. |
| **Finance** | Interest is recomputed from restated history, so a backdated entry across a month-end changes reported income; the general ledger needs reversing journals the core does not produce. |
| **Audit and AML** | Every record has both dates, so “what did we know on Day 5” is a query, and the log meets the five-year retention in Federal Decree-Law No. 10 of 2025. It does not record who backdated an entry or why. |
| **Context** | Under EU PSD2 Art. 87(3) a debit cannot be value-dated earlier than the debit itself, so the Part 1 backdated debit would be unlawful there as a customer payment. In a UAE bank, legitimate backdating is bank-side only: scheme clearing (T+1/T+2, carrying the transaction date), corrections and adjustments. |

> **The one control before go-live: a backdating horizon with maker-checker**
>
> The core refuses entries value-dated before the last closed day. Anything that must land behind it goes into an adjustments queue; a second person approves it with a reason code, and the approval is appended to the log.
>
> This bounds how much history can be rewritten, creates the audit trail an examiner asks for first, and is the condition under which automatic fee re-assessment is safe. The second control - linking each fee to the entry that caused it, so reversing the cause reverses the fee - is the failing test in Part 1 and item 4 in section 4.

## 3 Authorization lifecycle

In the model an authorization is a decision record whose state (declined, active, settled) is derived from whether a settlement references it. The only ending implemented in full is a matching settlement. The others:

| ENDING | REAL-WORLD CASE | WHAT I WOULD MANDATE |
|---|---|---|
| **Declined at request**<br>Today: recorded, no hold, never revisited | Not enough available funds (in practice also fraud rules). | Final, with a reason code; a later backdated credit does not re-open it. |
| **Duplicate request**<br>Today: rejected as an error | Network retry. | Return the original decision, no second hold. |
| **Never settled**<br>Today: the hold lives forever | Abandoned pre-authorization - hotel, car rental, fuel, a cancelled order. | Expire per scheme rules - Visa 7 days (31 for lodging, car rental, cruise), Mastercard 7 (30 for pre-authorizations); release the hold, record the expiry. |
| **Reversed by the merchant**<br>Today: no such event | Order cancelled before capture. | A new event; full reversal voids the hold, partial reduces it; only the merchant can originate it. |
| **Late presentment or no authorization**<br>Today: rejected (Part 1 criterion 4) | Stand-in processing while the issuer was down, offline chip approval, force-post. | Post it, flagged, so the ledger matches money the scheme has already settled, and send it to disputes; the remedy is a chargeback, not refusal. |

| ENDING | REAL-WORLD CASE | WHAT I WOULD MANDATE |
|---|---|---|
| **Settled for a different amount**<br>Today: partial releases the whole hold; over-settlement posts with a warning | Final bill below the estimate; tips above it. | Partial releases the remainder unless multi-capture was flagged; over-capture within the scheme's tolerance for that merchant type. |
| **Settlement later reversed**<br>Today: funds return, hold not reinstated | Refund or chargeback. | Same as today: a refund is a new credit, a chargeback its own workflow. |
| **Account frozen or closed**<br>Today: not modelled | Fraud freeze, legal hold, dormancy, closure. | A freeze blocks new authorizations but keeps existing holds; closure releases them and routes late clearings to disputes. |
| **Incremental authorization**<br>Today: same id is treated as a duplicate | Hotel stay extended. | Amend the existing hold and reset its expiry. |

## 4 What I cut and why

Sixteen simplifications, grouped. Each row: what was cut, why it stayed out of scope, and the production risk it defers.

| SIMPLIFICATION | WHY IT STAYED OUT | RISK IT DEFERS |
|---|---|---|
| **Storage and scale** | | |
| **In-memory lists, linear scans, one global sequence** | Persistence was out of scope. | Quadratic close; single writer; a crash loses the idempotency set, so a replay can double-post. |
| **Backdating unbounded** | The exercise depends on it. | Full-history restatement; covered by the horizon in section 2. |
| **Money model** | | |
| **Single-sided entries, no double-entry, no transfers** | The stream has none. | No debits-equal-credits check; nothing reconciles to the general ledger. |
| **Simple daily interest, positive balances only, one capitalisation; no debit interest, tiers, or a profit model for an Islamic window** | Spec wording. | Wrong for any real product; a restated accrual across a reporting period has no journal. |
| **HALF_UP, rounded daily then summed** | Spec wording. | Must match the customer agreement (the difference is fils). |
| **Only an AED fee configured** | The spec gives one. | An overdrawn BHD account produces an error, not a charge; needs per-currency configuration. |
| **Fees** | | |
| **No link from a fee to the entry that caused it, no fee reversal** | The spec says when to assess, never when to unwind; this is the deliberately failing test. | A reversed bank error leaves the customer paying. Fix: a trigger reference on each derived entry and a compensating fee reversal. |

| SIMPLIFICATION | WHY IT STAYED OUT | RISK IT DEFERS |
|---|---|---|
| **Debits never declined, no overdraft limit, flat fee with no cap** | The spec gives none. | Unlimited overdraft and fee stacking; production needs a limit per account and a cap per cycle. |
| **Authorizations** | | |
| **No hold expiry, merchant reversal, incremental authorization or multi-capture** | None in the stream. | Holds lock funds forever; a merchant void, the most common real ending, cannot be represented. |
| **Approvals never revisited after a backdated debit** | Card networks work this way. | The exposure sits with the bank; nothing flags “approved on a balance later restated”. |
| **No account lifecycle** | Out of scope. | Freeze and closure behaviour for holds is undefined. |
| **Time** | | |
| **Integer days, no calendar** | The brief's days are abstract. | Weekends (Saturday–Sunday here), holidays, cut-offs and Gulf-time day boundaries are undefined. |
| **Future value dates refused** | Keeps “balance as of day d” unambiguous. | Standing and scheduled payments cannot be represented. |
| **Input and errors** | | |
| **Structural faults stop the replay (precision, unknown account, duplicate id, bad amount)** | Right for a test harness. | One bad message halts production; needs a dead-letter queue. |
| **Rejections are only logged** | The brief asks for errors to be printed. | A real force-post never reaches disputes; settled money goes unaccounted. |
| **No operator, channel, reason code or external reference on entries** | The stream carries none. | Audit knows what and when, not who or why; no reconciliation to scheme files. |

In one line: the core is a query over an immutable log - right and slow. Production keeps the log and adds a horizon, links on derived entries, and the authorization events the card world actually sends.

References: CBUAE Consumer Protection Standards (Circular 8/2020) Art. 8 · CBUAE Regulations 29/2011 Annexure 2 (2018) · Federal Decree-Law No. 10 of 2025 · PSD2 Art. 87 · Visa/Mastercard hold timeframes. Drafted with Claude, as disclosed in the Part 1 WORKLOG.